

Indirect Space Charge Maths

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1 Bassetti-Erskine Formula

The Bassetti-Erskine formula for the electric field is given by:

$$\mathcal{E}(z) = E_x + iE_y = i\sqrt{\frac{2\pi}{\sigma_x^2 - \sigma_y^2}} \left[e^{-S^2/2} w\left(\frac{\bar{\omega}}{\sqrt{2(\sigma_x^2 - \sigma_y^2)}}\right) - w\left(\frac{\bar{z}}{\sqrt{2(\sigma_x^2 - \sigma_y^2)}}\right) \right]$$

with $z = x + iy$, $\omega = \frac{\sigma_y}{\sigma_x}x + i\frac{\sigma_x}{\sigma_y}y$ and $S^2 = \frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}$. $w(z)$ is the Faddeeva function related to the complex error function.

2 Circular Chamber

In 2D (using cgs units) an expression for the electric field from a source $\rho(\mathbf{x})$ is given

$$E(\mathbf{x}) = \int d^2\mathbf{x}' \frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x}' - \mathbf{x}|^2} \rho(\mathbf{x}')$$

Where the expression:

$$\mathbf{G}(\mathbf{x}, \mathbf{x}') = 2 \frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x}' - \mathbf{x}|^2}$$

Acts like a Green's function.

Using the complex electric field $\mathcal{E}(z) = E_x + iE_y$ and complex position coordinate $z = x + iy$, we can express this neatly as:

$$\mathcal{E}(z) = E_x + iE_y = 2 \int d^2z' \frac{\rho(z')}{\bar{z} - \bar{z}'}$$

with

$$\mathcal{G}(z, z') = \frac{2}{\bar{z} - \bar{z}'}$$

Indeed the Bassetti-Erskine formula is derived using this formulation.

A point charge at position $\mathbf{x}$ inside a circle of radius R (centred on the origin) produces a negatively charged image charge at the position $\mathbf{x}_i = \frac{R^2}{|\mathbf{x}|^2}\mathbf{x}$ or in the complex formulation $z_i = \frac{R^2}{\bar{z}}$

Using this we can construct a new Green's function for the image charge:

$$\mathcal{G}_i(z, z') = -\frac{2}{\bar{z} - \bar{z}'_i}$$

It can be shown

$$\begin{aligned} \frac{1}{\bar{z} - \bar{z}'_i} &= \frac{1 - \bar{z}'_i/\bar{z} + \bar{z}'_i/\bar{z}}{\bar{z}(1 - \bar{z}'_i/\bar{z})} = \frac{1}{\bar{z}} + \frac{\bar{z}'_i/\bar{z}}{\bar{z} - \bar{z}'_i} = \frac{1}{\bar{z}} + \frac{z_i/\bar{z}}{\bar{z}z_i/\bar{z}'_i - z_i} = \frac{1}{\bar{z}} - \frac{z_i/\bar{z}}{z_i - z'} \\ \mathcal{G}_i(z, z') &= \frac{-2}{\bar{z}} + \frac{z_i/\bar{z}}{z_i - z'} \end{aligned}$$

Substituting this back into our convolution integral:

$$\mathcal{E}_i(z) = \int d^2 z' \left(\frac{-2}{\bar{z}} + \frac{z_i/\bar{z}}{z_i - z'} \right) \rho(z') = \frac{-2}{\bar{z}} Q + \frac{z_i}{\bar{z}} 2 \int d^2 z' \frac{\rho(z')}{z_i - z'}$$

Where $Q = 1$ is the normalised total charge. The remaining integral on the right is simply the direct field formula except $\bar{z}$ and $\bar{z}'$ are replaced with z_i and z' . This is equivalent direct field evaluated at $\bar{z}_i$

$$\mathcal{E}_i(z) = \frac{-2}{\bar{z}} + \frac{z_i}{\bar{z}} \mathcal{E}_d(\bar{z}_i)$$

3 Rectangular Chamber

Poisson's equation ($\nabla^2 \phi = -4\pi\rho$) for a Gaussian source inside a PEC rectangle may be solved using a Fourier series.

We consider a Gaussian source at position (x_0, y_0) , with widths σ_x, σ_y inside a PEC rectangle with lengths L_x, L_y with its bottom left corner at the origin.

We first express the potential $\phi(x, y)$ and source $\rho(x, y)$ as sin Fourier series since this guarantees $\phi = 0$ at the boundary (the PEC boundary condition).

$$\phi(x, y) = \sum_{n,m} \phi_{nm} \sin(k_n x) \sin(k_m y)$$

$$\rho(x, y) = \sum_{n,m} \rho_{nm} \sin(k_n x) \sin(k_m y)$$

The Laplacian is given by:

$$\phi(x, y) = - \sum_{n,m} \phi_{nm} (k_n^2 + k_m^2) \sin(k_n x) \sin(k_m y)$$

This gives us the equation

$$\phi_{nm} = \frac{4\pi\rho_{nm}}{(k_n^2 + k_m^2)}$$

Our source has the form:

$$\rho(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{(x-x_0)^2}{2\sigma_x^2} - \frac{(y-y_0)^2}{2\sigma_y^2}\right)$$

$$\rho_{nm} = \frac{4}{L_x L_y} \iint \rho(x, y) \sin(k_n x) \sin(k_m y) dx dy := I_x I_y$$

where I_x are integrals over $\mathbb{R}$. Which is an approximation that the entire source is within the rectangle.

$$\begin{aligned} I_x &= \frac{2}{L_x \sqrt{2\pi}\sigma_x} \int dx dy \exp\left[-\frac{(x-x_0)^2}{2\sigma_x^2}\right] \sin(k_n x) \\ &= \frac{1}{iL_x \sqrt{2\pi}\sigma_x} \int dx dy \exp\left[-\frac{(x-x_0)^2}{2\sigma_x^2}\right] e^{ik_n x} - c.c. \\ &= \frac{1}{iL_x \sqrt{2\pi}\sigma_x} \int dx dy \exp\left[-\frac{x^2}{2\sigma_x^2} + \left(\frac{x_0}{\sigma_x^2} + ik_n x\right)x - \frac{x_0^2}{2\sigma_x^2}\right] - c.c. \\ &= \frac{1}{iL_x \sqrt{2\pi}\sigma_x} \exp\left[-\frac{x_0^2}{2\sigma_x^2}\right] \int dx dy \exp\left[-\frac{x^2}{2\sigma_x^2} + \left(\frac{x_0}{\sigma_x^2} + ik_n x\right)x\right] - c.c. \end{aligned}$$

Using known formula for Gaussian integrals

$$\begin{aligned} &= \frac{1}{iL_x} \exp\left[ik_n x_0 - \frac{1}{2}k_n^2 \sigma_x^2\right] - c.c. \\ &= \frac{2}{L_x} \exp\left[-\frac{1}{2}k_n^2 \sigma_x^2\right] \sin(k_n x_0) \end{aligned}$$

Therefore,

$$\rho_{nm} = \frac{4}{L_x L_y} \exp\left[-\frac{1}{2}(k_n^2 \sigma_x^2 + k_m^2 \sigma_y^2)\right] \sin(k_n x_0) \sin(k_m y_0)$$

Thus, altogether:

$$\phi_{nm} = \frac{16\pi}{L_x L_y} \frac{\exp\left[-\frac{1}{2}(k_n^2 \sigma_x^2 + k_m^2 \sigma_y^2)\right]}{k_n^2 + k_m^2} \sin(k_n x_0) \sin(k_m y_0)$$

Electric fields are expressed as gradients of the potential $\mathbf{E} = -\nabla\phi$

$$\begin{aligned} E_x(x, y) &= -\sum_{n,m} k_n \phi_{nm} \cos(k_n x) \sin(k_m y) \\ E_y(x, y) &= -\sum_{n,m} k_m \phi_{nm} \sin(k_n x) \cos(k_m y) \end{aligned}$$